

Reducing Hallucination in Retrieval-Augmented Generation Using Hybrid Vector–Graph Integration with Evidence Grounding

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Abstract- Large language models (LLMs) are intensely utilized to generate and analyze text, yet they also hallucinate and generate fluent texts that contradict accepted facts. Recent benchmark studies report hallucination rates ranging from 1.47% in clinical documentation to 69–88% in legal reasoning tasks, highlighting how reliability varies widely across domains. This proposal presents a retrieval technique that is a hybrid of semantic vector search and graph-based search that is more effective in providing contextual meaning in documents and the structure of relationships. The generated answers will be guided by retrieved evidence and supported by citation and abstention mechanisms in the case where there is inadequate evidence to look back on. The project will assess whether the method makes the results more reliable, less unsupported and the results of the LLM itself more credible than single-retrieval methods. Enhancing the quality of evidence grounding and quality of retrieval, the work is likely to enhance confidence in the work using LLM-based systems and implement them in a responsible manner when deciding based on documents.

Keywords - *Hallucination reduction, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Hybrid retrieval, Evidence grounding, Calibration.*

I. INTRODUCTION

The application of artificial intelligence is the generation, processing, and interpretation of information in education, healthcare, business, and enterprise activities. The systems using LLM, like ChatGPT, Gemini, and Claude, are used in document summarization, answering questions, and analytical help. But they cannot be so reliable, even as recent benchmarks show that the rates of hallucination are less than 2% in clinical summary tasks, through to over 80% in legal reasoning tasks, their output can be anti-factual [3], [4].

RAG seeks to address this issue by grounding model outputs in external, verifiable document sources rather than relying solely on internal training knowledge [1], [3]. Retrieval-augmented models such as Blended RAG and HybridRAG demonstrate improved contextual relevance and factual accuracy [1], [2]. Nevertheless, many systems rely on a single retrieval algorithm, often dense vector search, which may fail to capture critical associations among documents and facts [3], [7]. This motivates the integration of multiple retrieval methods to strengthen evidence grounding in document-based question answering.

II. PROBLEM

A. Problem Definition

It is anticipated that document-based question answering language models yield correct and evidence based answers based on reliable source documents, like reports, policies and enterprise knowledge base. As a matter of fact, these systems usually produce inconsistent or obsolete responses, and cannot enforce citation or abstention, as well as provide erratic results when queries are expressed differently. Those professionals and organizations that leverage AI-assisted services, e.g. enterprise search tools and AI assistants, in making high-stakes decisions are impacted by these limitations.

B. Significance of Problem

This issue is important as it is associated with the growth of the use of language models in the setting that handle large and often changing sets of documents. Poor decisions, compliance and operational risks and decreased trust in automated tools may occur due to poor accuracy or deficiency of supporting responses. When citation enforcement and consistency are weak, systems applied to platforms like Google Gemini, Microsoft Copilot and ChatGPT Enterprise might present hallucinated information with perceived authority. Although methods such as Retrieval-Augmented Generation (RAG), hybrid retrieval, reranking, and agentic approaches have been presented to enhance grounding, inaccuracies in quality of retrieval can still spread. This underscores the necessity to have more robust evidence-based principles to enhance reliability, transparency, and trust in document-grounded language model systems.

III. PRIOR WORK

RAG has become an important approach for improving the reliability of text generated by LLMs by grounding their responses in supporting documents. Prior research can be grouped into three areas: improving retrieval quality, reducing hallucinations using reliability-focused methods, and applying RAG techniques in real-world systems.

Early work on RAG mainly focused on improving retrieval accuracy so that the model receives relevant evidence.

Sawarkar et al. proposed Blended RAG, which uses a hybrid query strategy that combines dense vector search with sparse encoder indexing [1]. Their results show improved retrieval accuracy and ranking quality on datasets such as NQ and TREC-COVID. Building on this idea, Sarmah et al. introduced HybridRAG, which combines unstructured neural retrieval with structured knowledge graphs to improve contextual understanding and factual accuracy [2].

Later research shifted toward improving reliability and identifying hallucinations. Koç et al. proposed ReRAG, a feedback-based approach that adjusts retrieval settings such as chunk size and overlap using ideal question-answer examples [3]. This improves how well retrieved evidence matches the expected answers but depends on domain-specific data. Hu et al. proposed LRP4RAG, which uses Layer-Wise Relevance Propagation to analyze how retrieved tokens influence generated responses [4]. This helps with interpretability and hallucination detection.

Yadkori et al. suggested Conformal Abstention whereby the models can abstain in case of low confidence and offer statistical assurances on risk of hallucinations [5]. Chan et al. illustrated how prompt-engineered confidence scoring can enhance the accuracy and calibration of RAG-based technical support systems without retraining models and without altering system architecture [6]. Altogether, the current literature demonstrates improvement, but the practices are still rather isolated.

IV. PROPOSED METHODOLOGY

A. Plan

Through this project, a document-grounded question answering system will be designed, implemented, and evaluated to enhance the reliability of LLM outputs and decrease hallucinations and enhance evidence-based answers. The job shall be carried on through a systematic sequence of steps which will transform uploaded documents into a workable body of knowledge after which said body of knowledge will be utilized in responding to questions posted by users in a traceable manner. Fig 1 illustrates the end-to-end workflow of the proposed hybrid retrieval system. To start with, we shall gather and organize documents (e.g., PDFs or DOCX). The text will then be extracted out of each document but as much structure as possible preserved and the contents extracted out will then be cleaned to eliminate noise and formatting artifacts to allow the content to be successfully searched. The cleaned material will be then broken down into smaller and meaningful chunks to access pertinent content of the documents in order to answer a particular question effectively. Once prepared, the system will execute a hybrid retrieval strategy to find the most relevant evidence to a query. The response generation process will be developed to generate responses based on the evidence retrieved instead of unfounded assumptions of the model. The system will be assisted by citation and confidence rules to enhance trust and transparency. Lastly, the project shall assess

the performance of the system based on retrieval and reliability measures to determine how well evidence is selected and how effectively hallucinations are reduced. Such an approach will make sure that the plan of the follow-up is directly correlated to the objective of enhancing evidence-based LLM behavior on large document sets.

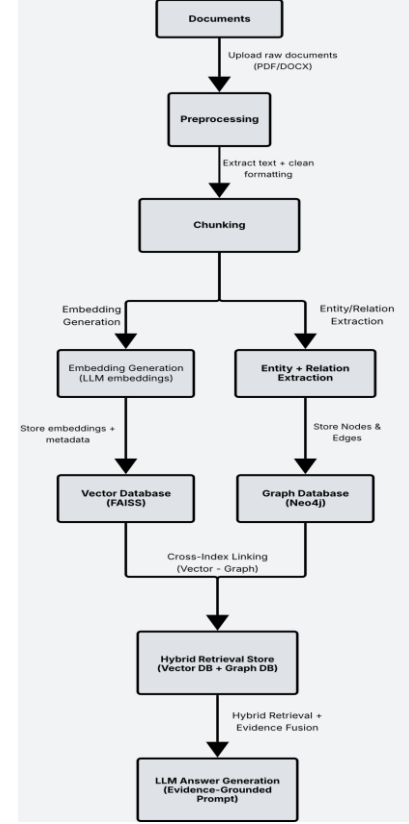


Fig. 1. Hybrid Vector-Graph RAG system architecture.

B. Challenges or Barriers

The main issues are accurate text extraction of complicated PDFs (many columns, scanned pages, tables), which can lessen the quality of retrieval. This will be dealt with by strong extraction with OCR failover and quality checks. Embedding large document corpora can be costly in terms of compute and memory; cost reduction will involve employing batch processing and embedding caching. Another difficulty is to build the graph store since the entities and relationships are to be properly extracted after reading the text; it will be controlled by the iterative extraction process, validation rules, and refining mechanisms.

C. Project Deliverables

The following deliverables will be submitted:

1. Working RAG Web Application: A functioning web-based RAG system that accepts document uploads and generates evidence-grounded answers with citations.

2. Evaluation Report: A report summarizing the evaluation methodology, metrics used, and results evaluating retrieval quality and hallucination reduction.
3. Tableau Dashboard: A Tableau dashboard that visualizes key evaluation results and performance metrics.
4. Deployment/Run Guide: A document explaining how to install dependencies, configure, and run the application.

REFERENCES

- [1] K. Sawarkar, A. Mangal and S. R. Solanki, "Blended RAG: Improving RAG (Retriever-Augmented Generation) Accuracy with Semantic Search and Hybrid Query-Based Retrievers," 2024 IEEE 7th International Conference on Multimedia Information Processing and Retrieval (MIPR), San Jose, CA, USA, 2024, pp. 155-161, doi: 10.1109/MIPR62202.2024.00031.
- [2] Sarmah, Bhaskarjit, Benika Hall, Rohan Rao, Sunil Patel, Stefano Pasquali, and Dhagash Mehta. "HybridRAG: Integrating Knowledge Graphs and Vector Retrieval Augmented Generation for Efficient Information Extraction." arXiv:2408.04948. Preprint, arXiv, August 9, 2024. <https://doi.org/10.48550/arXiv.2408.04948>.
- [3] R. Koç, M. K. Gürkan and F. T. Yarman Vural, "ReRag: A New Architecture for Reducing the Hallucination by Retrieval-Augmented Generation," 2024 9th International Conference on Computer Science and Engineering (UBMK), Antalya, Türkiye, 2024, pp. 961-965, doi: 10.1109/UBMK63289.2024.10773428.
- [4] Hu, Haichuan, Congqing He, Xiaochen Xie, and Qunjun Zhang. "LRP4RAG: Detecting Hallucinations in Retrieval-Augmented Generation via Layer-Wise Relevance Propagation." arXiv:2408.15533. Preprint, arXiv, June 27, 2025. <https://doi.org/10.48550/arXiv.2408.15533>.
- [5] Y. A. Yadkori, I. Kuzborskij, D. Stutz, et al., "Mitigating LLM Hallucinations via Conformal Abstention," arXiv:2405.01563, Apr. 4 2024. doi: 10.48550/arXiv.2405.01563.
- [6] W. T. Chan, K. Hung, R. Ho, and G. M.-T. Man, "Optimizing Confidence Scoring in RAG-Based LLM Chatbots for Technical Support Services: A Prompt Engineering Approach," 2025 IEEE 15th Symposium on Computer Applications & Industrial Electronics (ISCAIE), May 2025, pp. 540-545. doi: 10.1109/ISCAIE64985.2025.11081158
- [7] S. Knollmeyer, M. U. Akmal, L. Koval, S. Asif, S. G. Mathias and D. Großmann, "Document Knowledge Graph to Enhance Question Answering with Retrieval Augmented Generation," 2024 IEEE 29th International Conference on Emerging Technologies and Factory Automation (ETFA), Padova, Italy, 2024, pp. 1-4, doi: 10.1109/ETFA61755.2024.10711054.